• PCCOE INTERNATIONAL GRAND CHALLENGE 2026 • PUNE, INDIA •

• THEME : AI FOR CLIMATE CHANGE •

EVALADAPT

AI-Assisted Evaluation of Urban Heat Adaptation

Challenge Topic / Domain : Climate Intelligence & Environmental Analytics

TEAM NAME : FELIX

TEAM MEMBERS

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PROBLEM STATEMENT/IDEA

THE PROBLEM

Climate adaptation is being implemented, but its actual effectiveness is difficult to establish.

For interventions such as Cool Roof Programs, implementation records and climate observations alone cannot establish whether the intended outcome was achieved.

EXISTING APPROACH

Implementation Tracking
Where and when was the intervention implemented?

Climate Monitoring
What changed in measured conditions?

Project Evaluation
Did the intervention meet its target?

THE GAP

Evidence remains fragmented across key evaluation dimensions.

IMPACT | EQUITY | TIME | SPACE | COMMUNITY

Who benefited? Did effects persist? Did impacts spill over? What explains the difference between expected and observed outcomes?

WHY IT MATTERS

Underperformance, unequal benefits, declining effectiveness and unintended effects may remain difficult to identify.

↓

Implementation
Evidence?
Actual impact?
Who benefited?
What should change?

OUR IDEA — EVALADAPT

A unified post-implementation evaluation system that connects real-world evidence to an explainable Continue / Modify / Investigate decision.

EXPECTED → OBSERVED → ATTRIBUTED → DIAGNOSED → ADJUSTED

OBJECTIVES

Evaluate what changed → Understand why → Improve what comes next

01 EVALUATE

Did the intervention actually work?

- Expected vs Observed
- Impact / Counterfactual
- Who benefited?

02 DIAGNOSE

Why did outcomes differ from expectation?

- Temporal change
- Spatial effects & spillovers
- Uneven / unintended outcomes

03 IMPROVE

What should happen next?

- Continue / Modify / Investigate
- Evidence-based adjustment
- Re-evaluate with new evidence

SDGs aligned with EVALADAPT

SDG 13 — Climate Action - **Primary SDG** improves climate adaptation actions.

SDG 10 — Reduced Inequalities - Identifies who benefits, who is missed, and uneven adaptation outcomes.

SDG 11 — Sustainable Cities & Communities -Directly supports urban climate resilience

SDG 9 — Industry, Innovation & Infrastructure - AI enabled Data Analytics

EXPECTED IMPACT

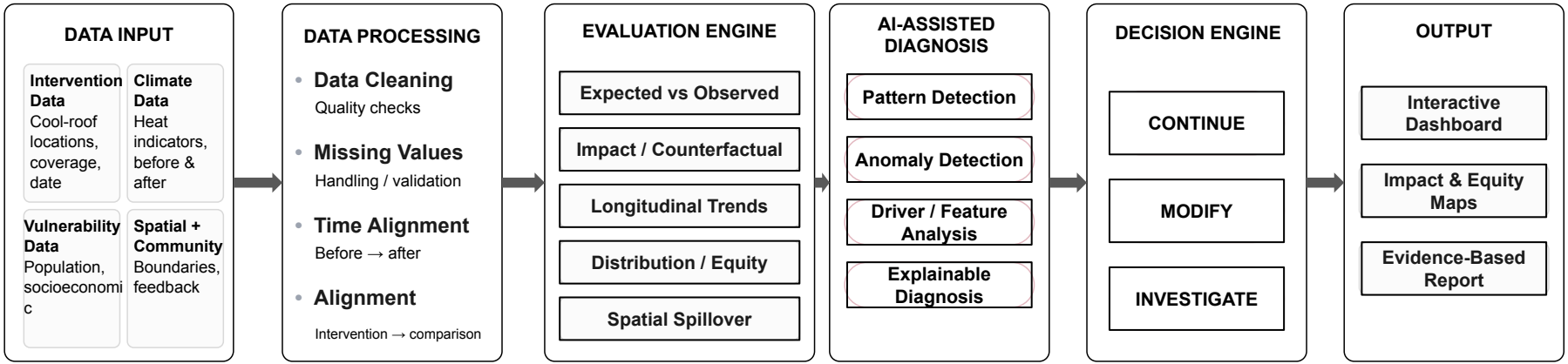
Better adaptation decisions	**More targeted investment**	**Visibility of unequal benefits**	**Continuous improvement**

END BENEFICIARIES

Governments / Policymakers • Cities & Implementers • Communities • Researchers & NGOs

PROPOSED SOLUTION/ARCHITECTURE

EVALADAPT



CONTINUOUS FEEDBACK

PLAN → ACT → OBSERVE → ATTRIBUTE → DIAGNOSE → ADJUST → RE-EVALUATE

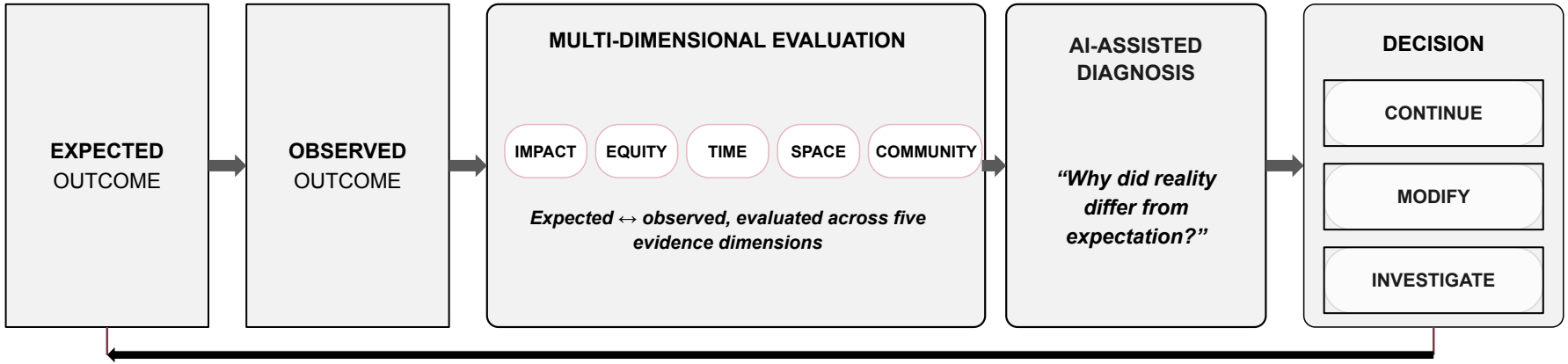
New observations + community feedback re-enter the evaluation cycle.

TECHNOLOGIES	MODULE INTERACTION	SCALABILITY
React • FastAPI • Python • Pandas • NumPy • Scikit-learn • GeoPandas • Plotly / Chart.js • SQLite	React UI → FastAPI → Evaluation Engine → AI Diagnosis → Decision Output	MVP: Urban Heat → Cool Roofs → Pune Future: Heat → Flood → Drought → Water → Agriculture

UNIQUE FEATURES/INNOVATION

FROM FRAGMENTED EVALUATION → CONTINUOUS ADAPTATION FEEDBACK

EVALADAPT integrates impact, equity, spatial, temporal and community evidence into one post-implementation evaluation workflow.



RE-EVALUATE — new observations and community feedback re-enter the cycle

01 OUTCOME-CENTRED

Implementation → Effectiveness

Evaluates whether the intervention achieved its intended outcome.

02 MULTI-DIMENSIONAL

Impact • Equity • Time • Space

Combines causal impact, distributional, longitudinal and spatial evidence.

03 EVIDENCE → ACTION

Diagnose → Decide

AI-assisted diagnosis converts results into Continue / Modify / Investigate.

04 CONTINUOUS FEEDBACK

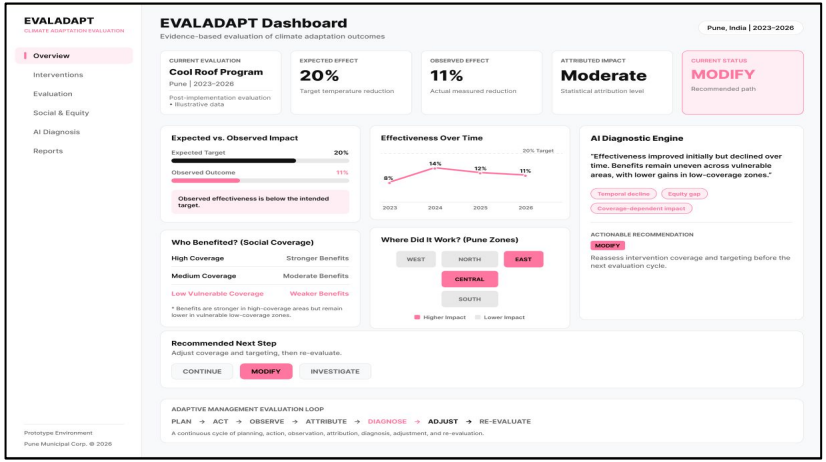
Evaluate → Adjust → Re-evaluate

New observations and community evidence feed the next evaluation cycle.

OUR INNOVATION The integration: connecting established evaluation methods and evidence sources into one operational, post-implementation feedback loop.

EXPECTED OUTCOME

01 — PROJECT DELIVERABLE



EVALADAPT Prototype

AI-assisted climate adaptation evaluation dashboard

02 — EXPECTED IMPACT

SHORT-TERM

- Identify underperforming interventions
- Detect uneven distribution of benefits
- Make evaluation evidence-driven
- Support targeted adaptation decisions

LONG-TERM

- Continuous adaptation learning
- Better allocation of climate resources
- Stronger accountability of interventions
- Evidence-based policy improvement

• Expected vs Observed Effectiveness	• Impact & Counterfactual Analysis
• Equity & Vulnerability Analysis	• Spatial & Temporal Trends
• AI-Assisted Diagnosis	• Continue / Modify / Investigate

03 — SCALING & FUTURE DEVELOPMENT

MVP

Urban Heat
↓
Cool Roofs — Pune

MORE ADAPTATION DOMAINS

Flood • Drought
Water • Agriculture

MORE GEOGRAPHIC SCALE

City → Region
→ State

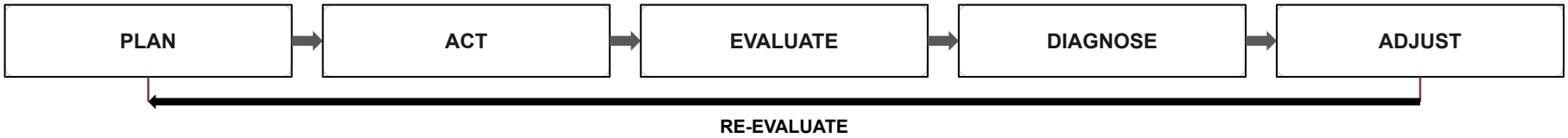
CONTINUOUS EVALUATION

Observe → Evaluate →
Diagnose → Adjust →
Re-evaluate

CONCLUSION

Adaptation is not complete when it is implemented.

Its effectiveness must be evaluated through real-world outcomes, distribution of benefits, and change over time.



01 | BETTER DECISIONS

Evidence-based adaptation choices

Evaluate expected vs observed outcomes and estimate intervention impact before deciding what happens next.

02 | EQUITY & UNINTENDED EFFECTS

Identify who benefits — and who is missed

Bring distributional effects, vulnerability and community evidence into adaptation evaluation.

03 | CONTINUOUS LEARNING

Evaluate → Adjust → Re-evaluate

Move beyond one-time assessment by using new observations and feedback to inform the next cycle.

REFERENCES - RESEARCH PAPERS

[1] Gaworek-Michalczenia et al. (2022) — *Evaluating the impact of adaptation interventions on vulnerability and livelihood resilience.*

[2] Shackleton et al. (2025) — *Human Development, Adaptive Capacity and Equity in Climate Adaptation Projects.*

[3] Dudney et al. (2025) — *A Causal Inference Framework for Climate Change Attribution in Ecology.*

[4] Musiyam et al. (2026) — *Can Machine Learning See Maladaptation?*

Additional literature reviewed across AI for climate adaptation, causal inference, equity, maladaptation and post-intervention evaluation.